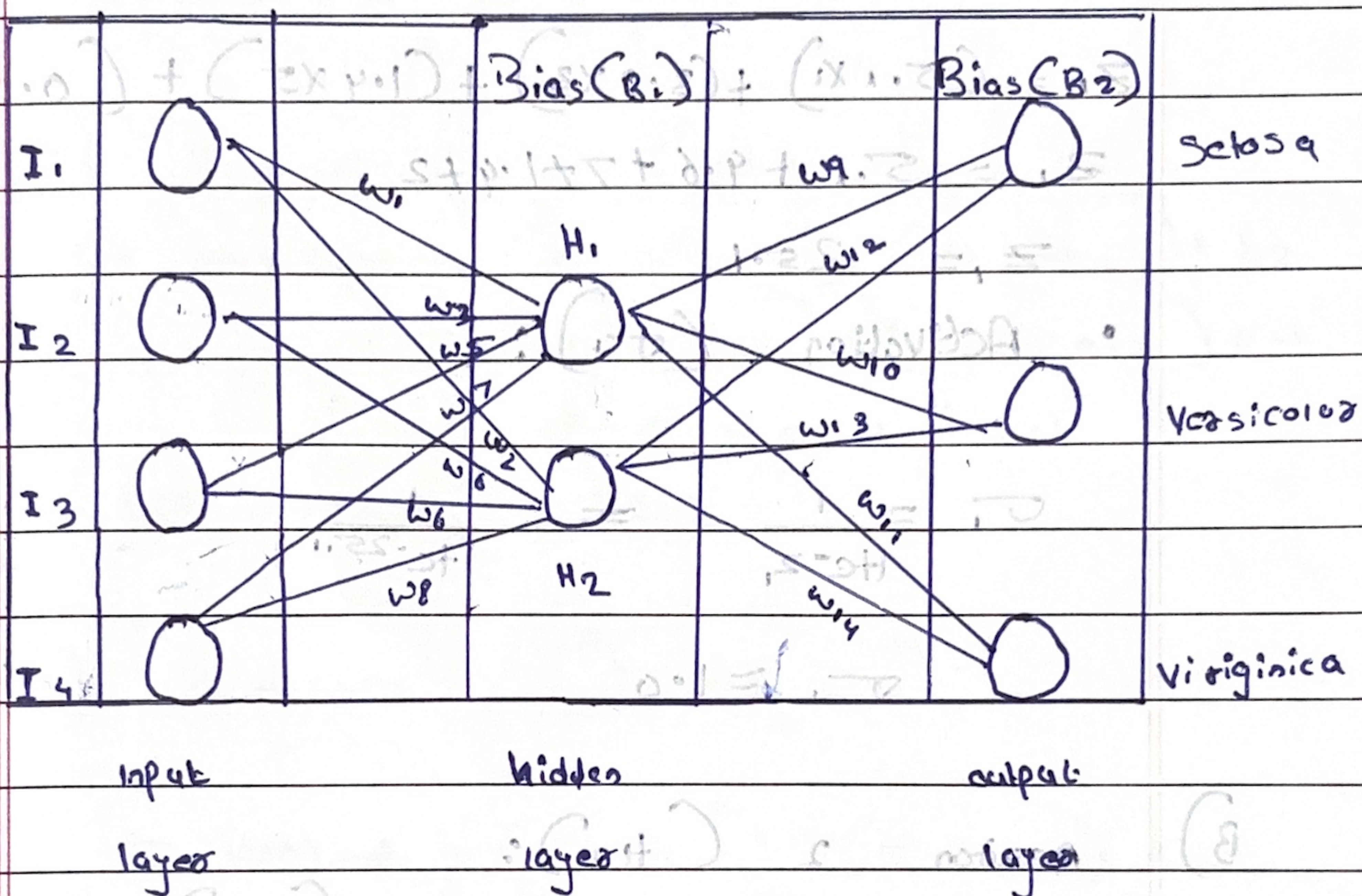


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Art

Mathematical Calculations Single - hidden - layer neural network



1) given Parameters

• input Vector: $X = [5.1, 3.2, 1.4, 0.2]$ • hidden weights (H_1): $w_1 = 1.25$ $w_3 = 3$ $w_5 = 5$ $w_7 = 7$ • hidden weights (H_2): $w_2 = 2$ $w_4 = 4$ $w_6 = 6$ $w_8 = 8$ • output weights (from H_1): $w_9 = 1.3$, $w_{10} = 0.5$ $w_{11} = 1.2$ • output weights (from H_2): $w_{12} = 0.5$, $w_{13} = 0.2$ $w_{14} = 0.4$ • biases $b_1 = 2$ $b_2 = [0.0, 0.0]$

2) hidden layer = Calculations on top

A) Neuron 1 (H_1):

• linear combination (z_1)

$$z_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b_1$$

$$z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2$$

$$z_1 = 5.1 + 9.6 + 7 + 1.4 + 2$$

$$z_1 = 25.1$$

• Activation (σ_1):

$$\sigma_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-25.1}}$$

$$\sigma_1 \approx 1.0$$

B) Neuron 2 (H_2):

• linear combination (z_2):

$$z_2 = x_1 w_5 + x_2 w_6 + x_3 w_7 + x_4 w_8 + b_2$$

$$z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.4 \times 6) + (0.2 \times 8) + 2$$

$$z_2 = 35.0$$

• Activation (σ_2):

$$\sigma_2 = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3) output layer calculation [Combining both neurons]

A) Set as a sigmoid ($\approx$ Sigmoid) instead

(z_1) $w_2 + b_1$

$$(4 \times 5) + (0.2 \times 7) + 2$$

$$\begin{aligned} \Sigma \text{Setosa} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1 \\ &= 1.3 + 0.5 + 0.1 \\ \Sigma \text{Setosa} &= 1.9 \end{aligned}$$

(B)

Vesicular ($\Sigma \text{Vesicular}$):

$$\begin{aligned} \Sigma \text{Vesicular} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1 \\ &= 0.5 + 0.2 + 0.1 \\ \Sigma \text{Vesicular} &= 0.8 \end{aligned}$$

C)

Virginia ($\Sigma \text{Virginia}$):

$$\begin{aligned} \Sigma \text{Virginia} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1 \\ &= 1.2 + 0.4 + 0.1 \\ \Sigma \text{Virginia} &= 1.7 \end{aligned}$$

4)

Final

Prediction

Output

Vector: $[1.9, 0.8, 1.7]$

Prediction

Setosa (highest Score 1.9)

[Combining both

a) input: